Find $f(1)$, $f(2)$, $f(3)$, $f(4)$ and $f(5)$ if $f(n)$ is defined recursively by $f(0)=3$ & for $n=0,1,2\ldots$

c) $f(n+1)=f(n)^2-2f(n)-2$

Ans: For, $f(1)$, $n=0$

$\therefore f(1)=f(0)^2-2f(0)-2$

$\boxed{f(1)=1}$

For, $f(2)$, $n=1$

$\therefore f(2)=f(1)^2-2f(1)-2$

$\boxed{f(2)=-3}$

For, $f(3)$, $n=2$

$\therefore f(3)=f(2)^2-2f(2)-2$

$=(-3)^2-2(-3)-2$

$\boxed{f(3)=13}$

for, $f(4)$, $n=3$

$\therefore f(4)=f(3)^2-2f(3)-2$

$=(13)^2-2(13)-2$

$\boxed{f(4)=141}$

for, $f(5)$, $n=4$

$\therefore f(5)=f(4)^2-2f(4)-2$

$=(141)^2-2(141)-2$

$\boxed{f(5)=19597}$

$f(1)=1$, $f(2)=-3$, $f(3)=13$, $f(4)=141$, $f(5)=19597$

d) $f(n+1)=3^{f(n)/3}$

for $f(1)$, $n=0$

$\therefore f(1)=3^{f(0)/3}$

$f(1)=3$

For, $f(2)$, $n=1$

$\therefore f(2)=3^{f(1)/3}$

$=3^{3/3}$

$f(2)=3$

for, $f(3)$, $n=2$

$\therefore f(3)=3^{f(2)/3}$

$=3^{3/3}$

$f(3)=3$

for $f(4)$, $n=3$

$f(4)=3^{f(3)/3}$

$=3^{3/3}$

$f(4)=1$

for $f(5)$, $n=4$

$f(5)=3^{f(4)/3}$

$=3^{3/3}$

$f(5)=3$

$f(1)=3$

$f(2)=3$

$f(3)=3$

$f(4)=3$

$f(5)=3$

find 7^{222} mod 11

$\Rightarrow 7^{111})^2 \Rightarrow 7^{55})^2 7)^2 \Rightarrow 7^{27})^2 7)^2 7)^2 \Rightarrow 7^{13})^2 7)^2 7)^2 7)^2$

$\Rightarrow \cancel{7)^6)} \Rightarrow 7^6)^2 7)^2 7)^2 7)^2 7)^2 \Rightarrow 7^3)^2)^2 7)^2 7)^2 7)^2 7)^2$

$\Rightarrow 7^1)^2 7)^2)^2 7)^2 7)^2 7)^2 7)^2$

$\therefore$ 7 mod 11 = ~~4~~ 7

$7^2 \times 7$ mod 11 = 2

2^2 mod 11 = 4

$4^2 \times 7$ mod 11 = 1

$1^2 \times 7$ mod 11 = 7

~~7^2 mod 11 = 5~~

~~So, 7^{222}~~

$7^2 \times 7$ mod 11 = 2

$2^2 \times 7$ mod 11 = 4

4^2 mod 11 = 5

$\boxed{\text{So, } 7^{222} \text{ mod } 11 = 5}$

$\gcd(123, 277)$

$= 277 \bmod 123$

$= 31$

$\gcd(123, 31)$

$= 123 \bmod 31$

$= 30$

$\gcd(30, 31)$

$= 31 \bmod 30$

$= 1$

$\gcd(30, 1)$

$30 \bmod 1$

$= \cancel{30}\ 0$

$\gcd(123, 277) = \cancel{30}\ 1$

$\gcd(1529, 14039)$

$= 14039 \bmod 1529$

$= 278$

$\gcd(1529, 278)$

$= 1529 \bmod 278$

$= 139$

$\gcd(139, 278)$

$= 278 \bmod 139$

$= 0$

$\gcd(1529, 14039) = 139$

If n is an integer and n^3+5 is odd, then n is even.

Let $P \to n^3+5$ is odd & $Q \to n$ is even

a) Proof by contraposition:

$\neg Q \to \neg P$

$\Rightarrow$ if n is odd then n^3+5 is even

Let, $n = 2k+1$

$n^3+5 = (2k+1)^3 + 5$

$= (2k)^3 + 3(2k)^2 \cdot 1 + 3 \cdot 2k \cdot (1)^2 + (1)^3 + 5$

$= 8k^3 + 12k^2 + 6k + 1 + 5$

$= 8k^3 + 12k^2 + 6k + 6$

$= 2(4k^3 + 6k^2 + 3k + 3)$

Let, $4k^3 + 6k^2 + 3k + 3 = m$

$\therefore n^3+5 = 2m$

[prooved)

b) Proof by contradiction :

$P \wedge \neg Q$

$\Rightarrow n^3+5$ is odd and n is odd

$\therefore n = 2k+1$

$n^3+5 = (2k+1)^3+5$

$= 8k^3+12k^2+6k+1+5$

$= 2(4k^3+6k^2+3k+3)$

Let, $4k^3+6k^2+3k+3 = m$

$n^3+5 = 2m$

[if n is odd then n^3+5 is even]

[proved]

Let $Q(x,y)$ be "x has sent an email message to y",
where x & y both domains consist of all students in class.

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a) $\exists x \exists y\, Q(x,y)$

$\Rightarrow$ At least one student has sent email to ~~at least~~ another student

b) $\exists x \forall y\, Q(x,y)$

$\Rightarrow$ There is at least one student who has sent an email to everyone in class.

c) $\forall x \exists y\, Q(x,y)$

$\Rightarrow$ Every student in class has sent an email to at least one student.

d) $\exists y \forall x\, Q(x,y)$

$\Rightarrow$ At least one student has received an email from every student in class.

e) $\forall y \exists x\, Q(x,y)$

$\Rightarrow$ Every student has received an email from at least one student.

f) $\forall x \forall y\, Q(x,y)$

$\Rightarrow$ Every student has received an email from every student in class.

Let $L(x,y)$ be "x Loves y", where domain for both x and y consists of all people in the world.

a) Everybody Loves Jerry

$\Rightarrow$ ~~$\forall x \exists$~~ $\Rightarrow \forall x\, L(x, \text{Jerry})$

b) Every body loves some body

$\Rightarrow \forall x \exists y\, L(x,y)$

c) There is somebody who loves every body.

$\Rightarrow \exists y \forall x\, L(x,y)$

d) Nobody Loves every body

$\Rightarrow \neg \exists x \forall y\, L(x,y)$

e) There is somebody whom Lydia does not Love.

$\Rightarrow \exists y \neg L(\text{Lydia}, y)$

F) There is somebody whom no one Loves.

$\Rightarrow \exists y \neg \forall x\, L(x,y)$

g) There is exactly one person whom everyone Loves.

$\Rightarrow$ ~~$\exists y$~~ Let unique person be z

$\Rightarrow \forall x\, L(x,z)$

h) There are exactly two people whom Lynn Loves.

$\Rightarrow$ Let two people be, a & b

$\Rightarrow L(\text{Lynn}, a) \wedge L(\text{Lynn}, b)$

i) Everyone loves himself or herself

$\Rightarrow \forall x\, L(x,x)$

j) There is someone who loves noone beside himself or herself.

$\Rightarrow \exists x\, L(x,x)$

C(x) is "x is a comedian" & F(x) is "x is funny" ::
domain consists of all people.

a) $\forall x(C(x) \rightarrow F(x))$

➢ if All are comedians then all are funny

b) $\forall x(C(x) \land F(x))$

➢ Everyone is a comedian and everyone is funny

c) $\exists x(C(x) \rightarrow F(x))$

➢ There is someone who, if they are a comedian then is funny

d) $\exists x(C(x) \land F(x))$

➢ There is someone who is a comedian and is funny.

Truth value of $(p \rightarrow q) \land (\neg p \rightarrow q)$

p	q	$p \rightarrow q$	$\neg p$	$\neg p \rightarrow q$	$(p \rightarrow q) \land (\neg p \rightarrow q)$
0	0	1	1	0	0
0	1	1	1	~~0~~ 1	1
1	0	0	0	1	0
1	1	1	0	1	1

Truth table of $(p \leftrightarrow q) \lor (\neg p \leftrightarrow q)$

p	q	$\neg p$	$p \leftrightarrow q$	$\neg p \leftrightarrow q$	$(p \leftrightarrow q) \lor (\neg p \leftrightarrow q)$
0	0	1	1	0	1
0	1	1	0	1	1
1	0	0	0	1	1
1	1	0	1	0	1

☐ Encrypt message "STOP GLOBAL WARMING" using shift cipher, with shift $k = 11$

⇒ "DEZA RWZMLW HLCXTYR"

☐ There are infinitely many primes.

⇒ Let $Q = P_1 . P_2 . P_3 . \dots . P_n$ [where Q is the complete list of prime numbers

Now, create a new prime number

⇒ $Q_n = (P_1 \cdot P_2 \cdot P_3 \cdot \dots P_n) + 1$

∴ In this new prime there is a remainder of 1.

∵ But in the original list there was no remainder.

So, Q_n itself is a new prime number

[proving the statement true]

☐ If n is a composite integer, then n has a prime divider less than or equal to $\sqrt{n}$.

⇒ Using Trial division,

∴ n is composite,

∴ then $n = a \times b$

⇒ Using proof by contradiction,

∴ suppose both factors are greater than $\sqrt{n}$

⇒ $a > \sqrt{n}$, $b > \sqrt{n}$

⇒ $a \cdot b > \sqrt{n} \times \sqrt{n}$

⇒ $ab > n$

[But this impossible, because $\boxed{ab = n}$]

[So statement is true]

Using Linear search

```
a=[1,3,4,5,6,8,9,11]

  find:=9
  location:=0
  For i:=0 to n do
      if a[i]= find then
          location= i
      end if
  end for
   return location
```

Using Binary search

```
array= [1, 2, 3, 4, 5, 6, 8, 9, 11]
   find :=9
   low :=0
   high:= length (array) -1
  while low <= high
        middle := floor ( (low+ high)/2)
            if array [middle] < find
                 low= midd+1
        else if array [middle] > find
                 high = middle -1
            else
          print "Found in index", middle
               stop
            end if
     end while
      Print "not found"
```

$a = [6, 2, 3, 1, 5, 4]$

```
for i:=1 to a-1
    for j:=1 to a-1
        if aj > aj+1, then interchange aj and aj+1
      end if
    end for
  end for
  print a
```